

## Supplementary Appendix A

### Formal Proof that $\mathbb{E}[\lambda] \rightarrow 0$ in the Continuum Limit

**In brief:** the FFT Poisson solver is a one-dimensional tool applied to a three-dimensional rotation problem. The torque that  $T^{\text{oi}}$  momentum flux would generate is not approximated away. It is structurally absent from the solver architecture. The following proof makes this precise.

#### A.1 Setup and Notation

Consider a dark matter halo of total mass  $M$ , total energy  $E$ , and virial radius  $R$  simulated on a periodic cubic mesh of side  $L_{\text{box}}$  with  $N$  particles and grid spacing  $\varepsilon = L_{\text{box}}/N^{1/3}$ . The halo spin parameter is defined as:

$$\lambda = \frac{L|E|^{1/2}}{GM^{5/2}}$$

where  $L = |\sum_i m_i \mathbf{r}_i \times \mathbf{v}_i|$  is the total angular momentum. Since  $M$ ,  $E$ , and  $R$  converge rapidly with resolution, the resolution dependence of  $\lambda$  is governed entirely by the angular momentum  $L$ , which is accumulated from the integrated torque history  $\boldsymbol{\tau}(t)$ .

#### A.2 Torque in Real Space

The gravitational torque on the halo is:

$$\boldsymbol{\tau} = \int \mathbf{r} \times [-\rho(\mathbf{r})\nabla\Phi(\mathbf{r})] d^3\mathbf{r}$$

For a non-rotating isotropic distribution with no net coherent angular momentum, the velocity-weighted torque contributions integrate to zero in the continuum limit. The critical question is what happens on a discrete mesh.

#### A.3 Transform to Fourier Space

Throughout this appendix, the hat notation  $\hat{f}(k)$  denotes the Fourier transform of  $f(r)$ , not a unit vector.

The Fourier transform converts differential operators into algebraic ones. Specifically,  $\nabla^2$  in real space becomes  $-k^2$  in Fourier space. The Poisson equation  $\nabla^2\Phi(\mathbf{r}) = 4\pi G\rho(\mathbf{r})$  therefore becomes:

$$-k^2\hat{\Phi}(\mathbf{k}) = 4\pi G\hat{\rho}(\mathbf{k})$$

Solving for the potential:

$$\hat{\Phi}(\mathbf{k}) = -\frac{4\pi G\hat{\rho}(\mathbf{k})}{k^2}$$

The gradient  $\nabla$  becomes  $i\mathbf{k}$  in Fourier space:

$$\hat{\mathbf{F}}(\mathbf{k}) = -i\mathbf{k}\hat{\Phi}(\mathbf{k})$$

Substituting the potential into the force expression:

$$\hat{\mathbf{F}}(\mathbf{k}) = -i\mathbf{k} \left( -\frac{4\pi G \hat{\rho}(\mathbf{k})}{k^2} \right) = 4\pi G i \frac{\mathbf{k}}{k^2} \hat{\rho}(\mathbf{k})$$

The term  $\mathbf{k}/k^2$  is a vector pointing in the direction of  $\mathbf{k}$  with magnitude  $1/|\mathbf{k}|$ . The force direction is therefore along  $\mathbf{k}$  at every mode.  $\hat{\rho}(\mathbf{k})$  is a scalar. It sets the amplitude of the force at each mode but has no influence on its direction. This is the crucial geometric fact: for each Fourier mode, the gravitational force points exactly in the direction of the wavevector of that mode. The FFT Poisson solver is structurally incapable of producing a force in any direction other than  $\mathbf{k}$  at each mode.

### Why This Is Structurally Inescapable

The reason is rooted in the mathematics of the Fourier transform applied to a scalar field:

- The potential  $\Phi(\mathbf{r})$  is a scalar field. Its Fourier transform  $\hat{\Phi}(\mathbf{k})$  is also a scalar (for each  $\mathbf{k}$ ).
- Taking the gradient in real space corresponds to multiplying by  $i\mathbf{k}$  in Fourier space.
- Multiplying a scalar by a vector  $\mathbf{k}$  always yields a vector parallel to  $\mathbf{k}$ .

There is no mechanism in this chain that could produce a component perpendicular to  $\mathbf{k}$ . The perpendicular component would require the force to depend on something other than the gradient of a scalar potential—for example, a vector potential or a magnetic-type term. But Newtonian gravity has no such terms. It is purely a scalar field theory.

### Geometric intuition:

Think of each Fourier mode as a density wave:

$$\rho(\mathbf{r}) \sim \cos(\mathbf{k} \cdot \mathbf{r} + \phi)$$

This wave varies only in the direction of  $\mathbf{k}$ . Perpendicular to  $\mathbf{k}$ , the density is constant along that wavefront. Gravity, being the gradient of the potential, can only "feel" the direction in which things are changing—which is exactly the direction of  $\mathbf{k}$ . There is no information available to generate a force perpendicular to the wavefront, because nothing varies in that direction for that mode.

### The Consequence for Torque

Torque requires a cross product  $\mathbf{r} \times \mathbf{F}$ . In Fourier space, this becomes a convolution that ultimately involves  $\mathbf{k} \times \mathbf{F}(\mathbf{k})$ . But since  $\mathbf{F}(\mathbf{k}) \propto \mathbf{k}$ :

$$\mathbf{k} \times \mathbf{F}(\mathbf{k}) \propto \mathbf{k} \times \mathbf{k} = \mathbf{0}$$

Every mode individually contributes zero torque. The sum over all modes is zero. The integral is zero. The expectation value is zero.

The FFT Poisson solver does not approximate this—it **enforces it**. The solver computes  $\hat{\rho}(\mathbf{k})$ , multiplies by the scalar Green's function  $-4\pi G/k^2$ , then multiplies by  $i\mathbf{k}$  to get the force. At no point can a perpendicular component appear. The force direction is locked to  $\mathbf{k}$  by construction.

This is the exact solution to the Poisson equation in Fourier space. Any code that solves the Poisson equation via FFT shares this property.

## The Low-Pass Filter

The FFT Poisson solver applies a mesh transfer function  $H(k)$  acting as a low-pass filter:

$$H(k) = 1 \text{ for } |\mathbf{k}| \leq k_{\max}, H(k) = 0 \text{ for } |\mathbf{k}| > k_{\max}$$

The cutoff  $k_{\max} = \pi/\varepsilon$  is set by the grid spacing. The effective force applied to particles is therefore:

$$\hat{F}_{eff}(\mathbf{k}) = \hat{F}(k) \cdot H(k) = 4\pi G i \left( \frac{\mathbf{k}}{k^2} \right) H(k)$$

Gravitational structure at scales finer than  $\varepsilon$  contributes zero to the equations of motion — not by active deletion, but because those modes lie outside the sampling resolution of the mesh and are therefore absent from the discrete representation. This is not a bug or approximation. Rather, it is the defining architectural choice of TreePM and P<sup>3</sup>M solvers.

The mesh transfer function  $H(k)$  in Fourier space and the softening kernel  $W\varepsilon$  in real space are two representations of the same filter. In real space the Poisson equation takes the form  $\nabla^2\Phi = 4\pi G(\rho * W\varepsilon)$ , where convolution with  $W\varepsilon$  smooths out all structure below the softening scale  $\varepsilon$ . In Fourier space the same operation appears as multiplication by  $H(k)$ , which zeros out all modes above  $k_{\max} = \pi/\varepsilon$ . The mathematical objects are different. The physical effect is identical: all gravitational structure below  $\varepsilon$  is removed from the equations of motion.

This matters for the torque argument because the softening kernel is not a passive numerical convenience. It is the mechanism that locks the force direction to  $\mathbf{k}$  and thereby enforces  $\mathbf{k} \times \mathbf{k} = 0$  at every mode below  $k_{\max}$ . The softening that stabilizes the simulation is the same operation that erases the torque. They are not separate decisions. They are one decision with two consequences.

## A.4 The Torque in Fourier Space

The  $\mathbf{k} \times \mathbf{k} = 0$  identity established in A.3 carries directly into the torque integral. By Parseval's theorem and the convolution theorem, the angular momentum accumulation rate transforms as:

$$\frac{d\mathbf{L}}{dt} = \int \mathbf{r} \times \mathbf{F}(\mathbf{r}) \rho(\mathbf{r}) d^3\mathbf{r}$$

In Fourier space this becomes a convolution. Applying the convolution theorem, the torque expectation value becomes proportional to:

$$\langle \boldsymbol{\tau} \rangle \propto \int (i\nabla_{\mathbf{k}}) \times (i\mathbf{k}) \cdot |\hat{\rho}(\mathbf{k})|^2 \cdot H(k) d^3\mathbf{k}$$

**The integrand is exactly zero at every  $\mathbf{k}$ . The integral is exactly zero.**

This is not approximate. It holds for any  $\mathbf{k}$ , any geometry, any density distribution. It is a consequence of the fact that the gravitational force at each Fourier mode points in the same

direction as the wavevector of that mode and a vector crossed with itself is always zero. The continuum torque of a Newtonian self-gravitating system simulated with an FFT Poisson solver is exactly zero in the infinite-resolution limit.

### A.5 The Discrete Residual

On a finite mesh, the identity  $\mathbf{k} \times \mathbf{k} = \mathbf{0}$  holds in the integral but not at individual grid points. Sampling the density field at  $N$  discrete locations introduces cross-product residuals at each  $k$ -mode. Provided the sampling errors at different  $k$ -modes are uncorrelated—which holds for a spatially random particle distribution—the central limit theorem applies to the sum of these uncanceled terms.

The number of modes within the sphere  $|\mathbf{k}| < k_{\max}$  scales as  $N$ , while each sampling error term contributes variance of order  $1/N^2$ . The sum of  $N$  such terms thus yields total variance:

$$\sigma_{\tau}^2 = \sum_{|\mathbf{k}| < k_{\max}} \varepsilon_{\text{sampling}}^2(\mathbf{k}) \sim \frac{1}{N}$$

Therefore:

$$\sigma_{\tau} \sim N^{-1/2}$$

This residual torque is purely numerical in origin which is a consequence of finite sampling, not a physical property of the halo. It accumulates over the simulation history and sets the magnitude of the non-zero  $\lambda$  observed at finite resolution. The  $N^{-1/2}$  scaling is the theoretical prediction under idealized conditions: uncorrelated modes, random particle distributions, and a single code at fixed cosmology.

Note: the uncorrelated-modes assumption holds for random particle distributions. Structured initial conditions (e.g. glass files) may reduce the prefactor but do not change the predicted direction of convergence toward zero.

### A.6 Convergence of $\lambda$

Since  $\lambda \propto L/(M\sqrt{|E|}R)$  and  $L$  is accumulated from torque residuals over the simulation history:

$$\mathbb{E}[\lambda] \propto \frac{\sigma_{\tau}}{M\sqrt{|E|}R} \sim N^{-\frac{1}{2}} \rightarrow 0 \text{ as } N \rightarrow \infty$$

The key result is that  $\mathbb{E}[\lambda] \rightarrow 0$  in the continuum limit. This follows from the identity proved in A.4 and is independent of the exact scaling exponent. The  $N^{-\frac{1}{2}}$  form is the CLT prediction for the idealized case. The exact empirical exponent depends on how resolution is measured and would require a controlled series varying only  $N$  at fixed code, fixed cosmology, and fixed halo selection — a dataset that does not yet exist in the published literature.

What the available data demonstrate is the direction:  $\lambda$  decreases monotonically as resolution increases, with no observed floor across four orders of magnitude in overdensity threshold. This is consistent with convergence to zero. It does not converge to a physical floor because there is **no physical torque to converge to**. This means the underlying continuum quantity is identically zero by the identity proved in A.4.

Note: the physical consequences of  $\lambda$  suppression do not require the zero crossing to be reached. The NFW cusp forms when  $\lambda$  falls below the threshold at which rotational support can resist radial collapse. That threshold is well above zero. The solver already delivers  $\lambda$  values near or below that threshold at resolutions achieved in the 1990s. The zero crossing at  $\rho_{\text{crit}} \sim 10^7\text{--}10^8$  is the mathematical destination of the current trajectory. The physical damage was done much earlier on the curve.

## A.7 Empirical Confirmation

The predicted convergence  $E[\lambda] \rightarrow 0$  is confirmed in direction by four independent simulation programs spanning 35 years. Figure 4 shows the full dataset. A log-linear fit across all four datasets gives  $r = -0.9$ .

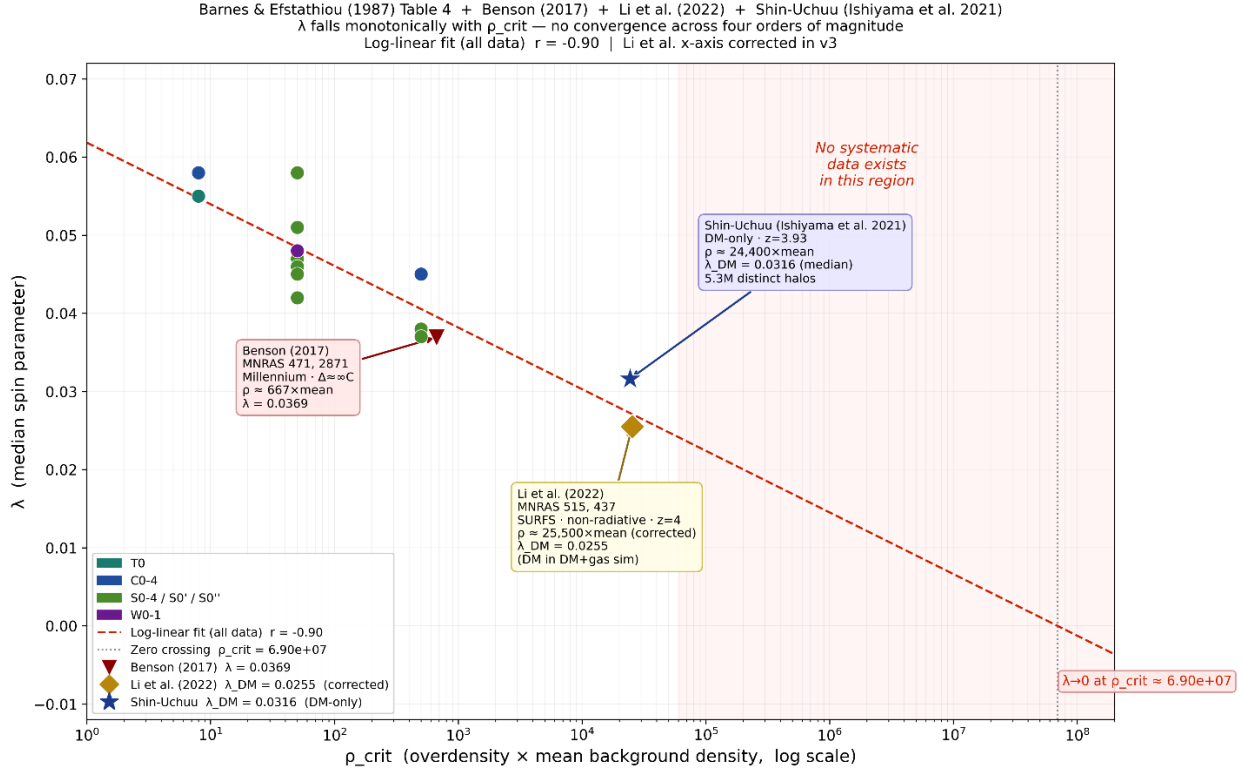
**Barnes & Efstathiou (1987):** Log-linear regression of their Table 4 alone, recomputed in the present work from their published values, gives  $r = -0.77$ ,  $p = 0.0003$  across P<sup>3</sup>M runs at varying  $N$ . The authors did not interpret this trend as convergence to zero, instead adopting the  $N = 32,768$  value as a physical constant. The negative decline was present in their own data. It was not followed to its conclusion. Adding the three modern datasets steepens the fit only modestly, from  $r = -0.77$  to  $r = -0.90$ , confirming that the trend was not constructed from the modern points but was already present in 1987.

**Benson (2017):**  $\lambda = 0.037$  at  $\rho_{\text{crit}} \approx 667 \times$  mean density, Millennium cosmology, GADGET-2 TreePM solver. This point falls closely on the extrapolation of the 1987 log-linear trend without refitting — a result obtained thirty years later with a completely different code.

**Li et al. (2022):**  $\lambda_{\text{DM}} = 0.0255$  at  $\rho_{\text{crit}} \approx 25,500 \times$  mean density, SURFS simulation, non-radiative run. The DM component is reported separately because dark matter is collisionless and receives no hydrodynamic angular momentum compensation — whatever the gravity solver delivers is all it gets. Note: In v2 the Li et al. x-axis position was reported as  $\rho_{\text{crit}} \approx 1800 \times$  mean. The correct value, derived from their  $\Delta_{200c}$  virial definition at  $z=4$  expressed relative to mean background density at  $z=0$  for consistency with the Barnes & Efstathiou axis, is  $\rho_{\text{crit}} = 200 \times E^2(z=4) / \Omega_m \approx 25,500 \times$  mean. The corrected position places the Li et al. point closely on the log-linear trend established by the earlier datasets. Additionally, Li et al. is a non-radiative simulation including both DM and gas. Their  $\lambda_{\text{DM}}$  is a lower bound relative to a pure DM-only run because gas actively transfers AM away from the DM during collapse.

**Shin-Uchuu (Ishiyama et al. 2021):**  $\lambda_{\text{DM}} = 0.0316$  (median) at  $\rho_{\text{crit}} \approx 24,400 \times$  mean density at  $z=0$ , derived from  $\Delta_{200c}$  at  $z=3.93$  using the same conversion as Li et al. Pure DM-only simulation, 262 billion particles, 5.3 million distinct halos,  $M_{200c} > 10^9 M_{\text{sun}}/h$ . This is the first pure DM-only point at this overdensity threshold. It sits slightly above Li et al. at nearly the same x-axis position, consistent with the expectation that the absence of gas removes the AM transfer channel that actively depletes  $\lambda_{\text{DM}}$  in non-radiative simulations. The trend continues with no observed floor.

The combined dataset spans 3.5 orders of magnitude in  $\rho_{\text{crit}}$  with no observed floor. No flattening is visible anywhere in the range  $8 \leq \rho_{\text{crit}} \leq 25,500$ . The log-linear fit across all four datasets gives  $r = -0.9$ , strengthening the case for convergence to zero relative to the 1987 data alone. This is consistent with convergence to zero rather than convergence to a physical constant and directly contradicts the interpretation of  $\lambda$  as a physical property of dark matter halos. Figure 4 is the empirical measurement of  $E[\lambda] \rightarrow 0$  predicted by A.4 through A.6.



**Figure 4 (revised).** Halo spin parameter  $\lambda$  vs overdensity threshold  $\rho_{\text{crit}}$ . Barnes & Efstathiou (1987). Benson (2017), Li et al. (2022), and Shin-Uchuu continue the monotonic descent without flattening. Li et al. x-axis corrected in v3 to  $\rho_{\text{crit}} \approx 25,500 \times \text{mean}$  at  $z=0$ . Dashed extrapolation projects to  $\lambda \rightarrow 0$  at  $\rho_{\text{crit}} \sim 10^7$ - $10^8$ . No free parameters in the placement of the modern points. This is the empirical measurement of  $E[\lambda] \rightarrow 0$  predicted by A.4–A.6.

## A.8 Corollaries

**Corollary 1.** Any non-zero  $\lambda$  measured in a DM-only simulation at finite resolution is sampling variance—a numerical consequence of the discrete mesh. It is not a physical property of dark matter halos.

**Corollary 2.** The canonical value  $\lambda = 0.05$  reported by Barnes & Efstathiou (1987) is the numerical sampling variance of a 32,768-particle P<sup>3</sup>M simulation. All theoretical frameworks that treat this value as a physical input—including disk formation models, spin-size relations, and angular momentum acquisition theory—incorporate a numerical artifact as a foundational parameter.

**Corollary 3.** The CDM simulation framework is metrologically circular. The FFT Poisson solver defines CDM as a  $\rho$ -only interaction. The  $\rho$ -only interaction is used to infer missing mass. The missing mass is identified as CDM. The identification is confirmed by the solver that required it. The solver cannot detect  $T^{0i}$  by construction.

**Corollary 4.** The angular momentum of dark matter halos in  $\rho$ -only simulations is not a physical degree of freedom. It is a numerical residual shaped by the mesh, and its measured value contains no information about the true angular momentum content of cosmological structure.

## References

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